

1. Describe the purpose of <Link> component in React Router. How does it differ from using an anchor (<a>) tag for navigation?

Answer:

The <Link> component in React Router is used for **client-side navigation** within a single-page application (SPA). It allows users to navigate between different routes **without causing a full page reload**, preserving the application's state and providing a smoother user experience.

In contrast, a traditional HTML <a> tag triggers a **full page reload** every time it's clicked, because it sends a request to the server for a new page. This breaks the SPA behavior.

Key Differences:

- <Link> uses React Router's internal routing system (JavaScript-based navigation).
- <a> tag sends a new HTTP request and reloads the page.
- <Link> maintains the app state, <a> does not.

Example:

```
<Link to="/about">About Us</Link> // React Router navigation
<a href="/about">About Us</a> // Traditional navigation (full reload)
```

2. What are route guards in React Router? How can you implement route guards to restrict access to certain routes based on user authentication?

Answer:

Route guards are mechanisms used to **restrict or control access** to specific routes in an application. They ensure that only authenticated or authorized users can access certain pages (for example, dashboard or admin pages).

In React Router, you can implement route guards by **wrapping protected routes** in a component that checks authentication before rendering.

Example Implementation:

```
import { Navigate } from "react-router-dom";
```

```
const ProtectedRoute = ({ isAuthenticated, children }) => {
  if (!isAuthenticated) {
    return <Navigate to="/login" />; // redirect to login if not logged in
  }
  return children; // render protected component if logged in
};
```

```
// Usage
<Route
  path="/dashboard"
  element={<ProtectedRoute isAuthenticated={userLoggedIn}><Dashboard /></ProtectedRoute>}
/>
```

Explanation:

- If the user is not authenticated, they're redirected to the login page.
 - If authenticated, the protected component (Dashboard) renders normally.
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3. What is the purpose of the <Switch> component in React Router? Provide an example scenario where <Switch> is beneficial.

Answer:

In older versions of React Router (v5 and below), the <Switch> component ensures that **only the first matching route** among multiple defined routes is rendered.

Without <Switch>, React Router renders **all routes** that match the current URL, which can cause multiple components to display at once.

Example Scenario:

```
<Switch>  
  <Route path="/about" component={About} />  
  <Route path="/contact" component={Contact} />  
  <Route path="/" component={Home} />  
</Switch>
```

If the current path is /about, only the About component will render — even though / also matches /about. This is beneficial when defining **mutually exclusive routes** in an application.

Note: In React Router v6, <Switch> has been replaced with <Routes> which behaves similarly but uses element instead of component.